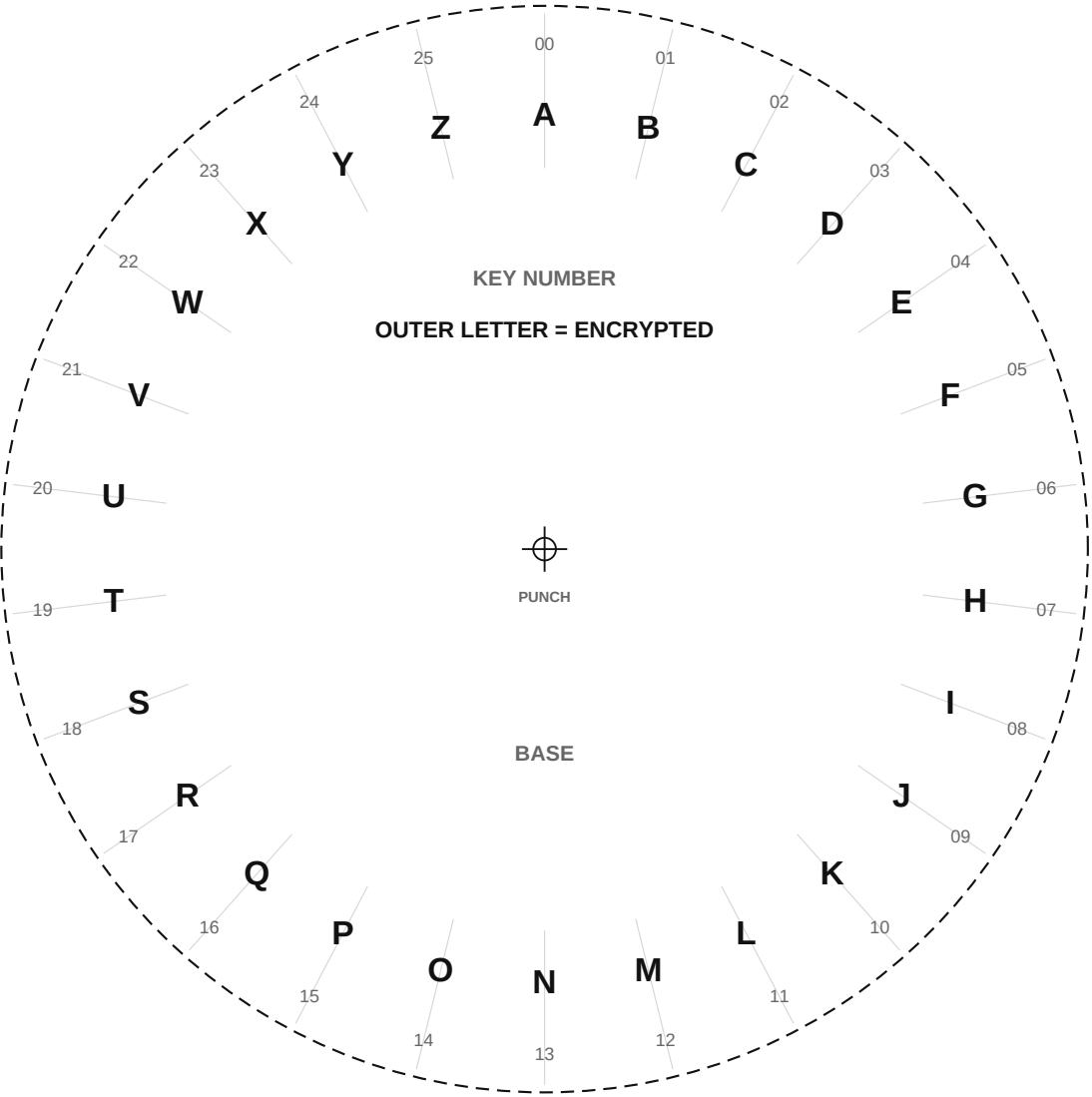


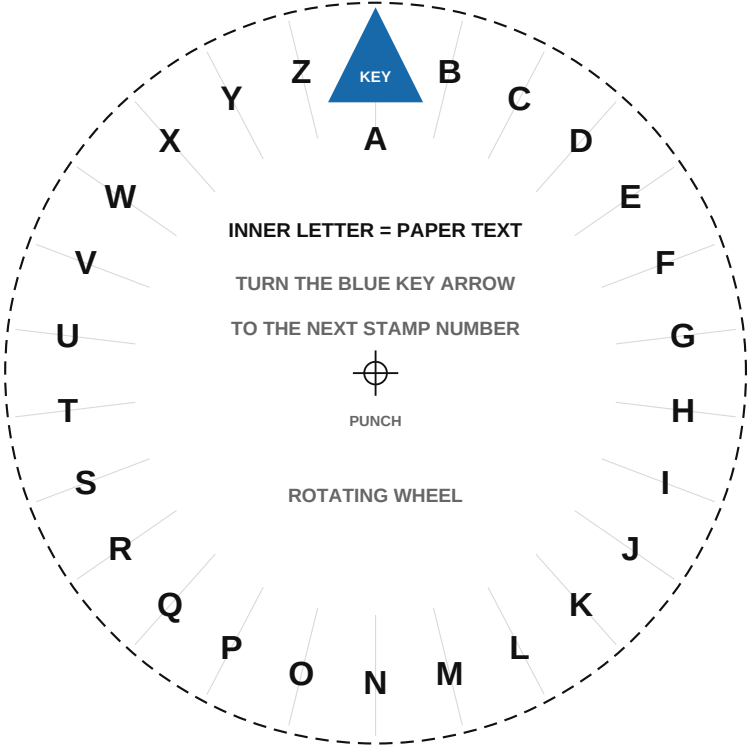
# PAPER CIPHER WHEEL - REAL PRINT TEMPLATE

Print at 100% / Actual Size. Do not use Fit to Page.



## ASSEMBLY

1. Cut out both dashed circles.
2. Stack them with the center marks matched.
3. Join the centers with a brass paper fastener.



This line must measure exactly 100 mm

# USE THE WHEEL

Plaintext stays on paper. Only encrypted letters go into the phone.

## SEND - paper text to encrypted text

1. Use the next unused number on your stamp strip.
2. Turn the blue KEY arrow to that number.
3. Find your paper letter on the INNER wheel.
4. Copy the matching OUTER letter to the phone.
5. Cross out that stamp number. Repeat.

## READ - encrypted text to paper text

1. Use Bob's matching unused stamp number.
2. Turn the blue KEY arrow to that number.
3. Find the received letter on the OUTER ring.
4. Copy the matching INNER letter onto paper.
5. Cross out that stamp number. Repeat.

## TRY IT - HI must become KJ

|      |         |         |         |         |         |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------|---------|---------|---------|---------|---------|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| TEST | 1<br>03 | 2<br>01 | 3<br>20 | 4<br>04 | 5<br>17 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|------|---------|---------|---------|---------|---------|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

First try: paper H I      key 03 01      encrypted result K J

Longer check: paper H E L L O      key 03 01 20 04 17      result K F F P F

## BLANK 26-NUMBER STAMP STRIPS - numbers must be 00 through 25

|         |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| STAMP 1 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

|         |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| STAMP 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

|         |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| STAMP 3 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

|         |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| STAMP 4 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

|         |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| STAMP 5 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

## NON-NEGOTIABLE SECURITY RULES

- A stamp must be truly random, Alice and Bob must hold matching copies, and every number may be used only once.
- For 2-10 stamps, run the whole message through the wheel once per stamp. Bob reverses each pass on paper.
- This wheel only performs the letter math. It does not safely create, share, authenticate, or store the stamps.